

Adaptive Attention Pruning for Real-Time Neural Inference on Edge Hardware*

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Abstract. Deploying Transformer-based architectures on resource-constrained edge computing hardware presents severe challenges due to the quadratic computational complexity of self-attention mechanisms. Existing compression methods, such as static pruning and post-training quantization, often fail to dynamically adapt to varying input complexity, leading to sub-optimal latency-accuracy trade-offs. In this paper, we introduce **AP-Edge**, an adaptive attention pruning framework tailored for real-time inference on edge processors. AP-Edge computes input-dependent dynamic sparsity masks by evaluating query-key magnitude distributions prior to full matrix multiplication. Our experimental evaluation on embedded accelerators demonstrates that AP-Edge achieves up to a $2.4\times$ speedup over unpruned baselines while preserving 98.6% of baseline accuracy across standard benchmarks.

Keywords: Edge Computing · Neural Pruning · Efficient Transformers · Real-Time Inference · Adaptive Sparsity.

1 Introduction

Transformer architectures have established state-of-the-art performance across computer vision, natural language processing, and spatial perception domains. However, their deployment on micro-controllers and embedded systems remains bottlenecked by strict memory bandwidth and energy consumption constraints [3]. The primary compute burden stems from the multi-head self-attention (MHSA) module, where sequence length N yields an $\mathcal{O}(N^2)$ time and spatial overhead.

To mitigate these resource demands, model compression techniques such as magnitude-based weight pruning and post-training quantization have been widely studied. While effective for feed-forward layers, static pruning cannot

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